

INDIAN STATISTICAL INSTITUTE, KOLKATA

An Analysis of Industrial Production Growth in India from 1994 to 2024

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June 2, 2026

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Acknowledgements

The author extends heartfelt gratitude to Prof. Ayanendranath Basu for providing the opportunity to undertake this project, and for his constant guidance and support. Sincere thanks are also due to Prof. Samarjit Das, Professor, Economic Research Unit at the Indian Statistical Institute, for his invaluable support, guidance, and assistance in obtaining the dataset.

I would also like to thank my colleagues and friends for their encouragement and constructive feedback, which has motivated me throughout this journey. Finally, I extend my appreciation to the academic community at the Indian Statistical Institute, Kolkata, for providing a conducive learning environment.

1 Introduction

Industrial production growth serves as a crucial indicator of a nation's economic progress, reflecting shifts in industrial output, employment, and investment patterns. In India, the period from 1994 to 2024 has witnessed significant transformations driven by economic liberalization, technological advancements, and policy reforms.

The Index of Industrial Production measures the quantum of changes in the industrial production in an economy and captures the general level of industrial activity in the country. Therefore, it can be used as a measure for short term industrial production growth in an economy.

In this project, we attempt to analyse the growth of industrial production in India from 1994 to 2024, using methods from Time-Series Analysis. We analyse the monthly IIP data from April 1994 to December 2024, and attempt to find the trend and the seasonal variations in the data.

1.1 Index of Industrial Production

The **Index of Industrial Production (IIP)** is a short term indicator of industrial performance. It measures changes over time in the volume of work done in various sectors of industry, related to the production of commodities, excluding agriculture and services. IIP is considered to be one of the lead indicators for short-term economic analysis because of its strong relationship with economic fluctuations in the rest of economy. Most of services, like transport, storage, communication, real estate, insurance and banking are industry dependent and are considerably influenced by industrial performance. The scope of the IIP as recommended by the United Nations Statistical Office (UNSO) includes mining, manufacturing, construction, electricity, gas and water supply. However, due to constraints of data availability, the IIP compiled in India has excluded construction, gas and water supply sectors.

1.2 Calculation of IIP

Index of Industrial Production is compiled and published every month by Central Statistics Office (CSO) of the Ministry of Statistics and Programme Implementation with a time lag of six weeks from the reference month. As a volume index, IIP is defined as the ratio of the volume of commodities produced within a specified group of industries in a given time period to the volume produced in the same group of industries in a specified base period. It is compiled as a simple weighted arithmetic mean of production relatives by using Laspeyre's formula. The index at time t is calculated as,

$$L_t = \frac{\sum_{i=1}^n R_i W_{i0}}{\sum_{i=1}^n W_{i0}} \times 100$$

where,

- W_{i0} is the weight of the i^{th} item in the base year
- $R_i = \frac{P_{it}}{P_{i0}}$, is the production relative of the i^{th} item, the ratio of the production of the i^{th} item in the period t , given by P_{it} and the production of the i^{th} item in the base period, given by P_{i0} .

2 Data Preprocessing

2.1 Linking Factors

The Central Statistical Organization (CSO) makes periodical revisions of the base period and weighting diagram of the IIP by reviewing the coverage of items and industries and by improving, as far as practicable, the technique used in the construction of IIP and by shifting the base to a recent period. This is required to account for the change in the structure of the industrial sector over time, so as to measure the real growth in the industrial sector.

This necessitates the use of **linking factors**, which are multipliers which help in comparing series with different base periods.

2.2 Calculation of Linking Factors

There are three methods by which linking factors may be calculated:

1. Arithmetic Conversion Method: In this method, the relationship between indices in old series(y) and new series(x) is assumed to be linear, i.e. $y = cx$, where c is the conversion factor given by $c = \frac{\bar{y}}{\bar{x}}$, where $\bar{y}$ and $\bar{x}$ are the average values of the indices in the two series for the year chosen as the base period of the new series. In practice, we take $\bar{x} = 100$.
2. Ratio Method: In this method, the month-wise ratio of new indices and old indices is worked out, and their average is taken as the linking factor.
3. Regression Method: We assume a relation $y = \alpha + \beta x$ between the common time period in the two series, and then perform ordinary least squares regression of y on x to obtain estimates for α and β . Using these, we can convert one series to other series.

Between 1994 and 2024, the base periods adopted were 1993-94, 2004-05, and 2011-12. In accordance with government practices, we have used the arithmetic conversion method to calculate the linking factors between the three series. The linking factors are in the following table:

Base Year	Linking Factor
2004-05 to 1993-94	2.111
2011-12 to 2004-05	1.703

3 Data Visualization

3.1 Visual Inspection of Time Series

We divide the series with older base year by the appropriate linking factor to obtain the combined series. Thus, we first divide the series with base year 1993-94 by 2.111 and the combined series (including the adjusted series with base year 1993-94) with base year 2004-05 by 1.703 to obtain our final series, which we shall analyse.

Visual inspection of the series suggests the following :

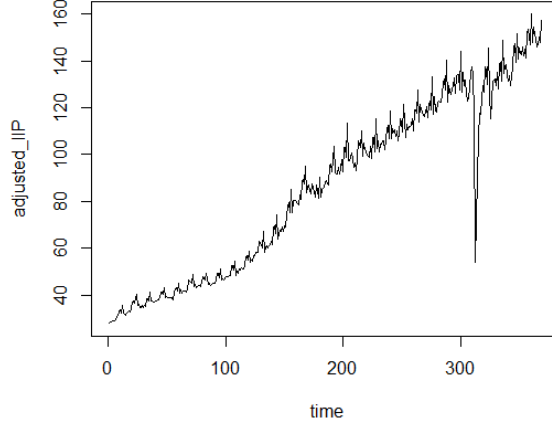


Figure 1: Time Series Showing Adjusted IIP from 1993-94 to 2024-25

1. **Overall Upward Trend** : There is a clear long term increase in the values from 1994-95 to 2024-25.
2. **Seasonality** : The periodic spikes and dips suggests the presence of seasonal components in the data.
3. **Heteroscedasticity** : The variance appears to increase with the increase in level, suggesting that a multiplicative model may be better at explaining the variation in the data.
4. **Presence of Anomaly** : There is a sharp dip in the series, just beyond $t = 300$. This corresponds to the year 2020, which suggests that the dip was most likely caused due to the COVID-19 pandemic. The anomaly seems to be a point anomaly, as the series clearly returns to its upward trajectory.

4 Model Decomposition

4.1 Model Selection

We need to determine if the series follows an additive model or a multiplicative model before proceeding with our analysis.

Additive model :

$$Y_t = m_t + s_t + \epsilon_t$$

Multiplicative model :

$$Y_t = m_t \cdot S_t \cdot \epsilon_t$$

where m_t denotes trend component, s_t denotes seasonal component, ϵ_t denotes irregular variation, at time period t . We decompose the series into the additive model and the multiplicative model by the following method:

1. In each case, we first determine the trend component using a moving average. We use a symmetric window with equal weights to estimate trend.

2. The estimated trend is then removed from the time series. We now compute the seasonal figure by averaging for each time unit over all periods. The seasonal figure is then centered.
3. The irregular variation is determined by removing the trend and seasonal figure (recycled as needed) from the original time series.

We obtain the decompositions as shown in Figure 2 for the additive model and as shown in Figure 3 for the multiplicative model.

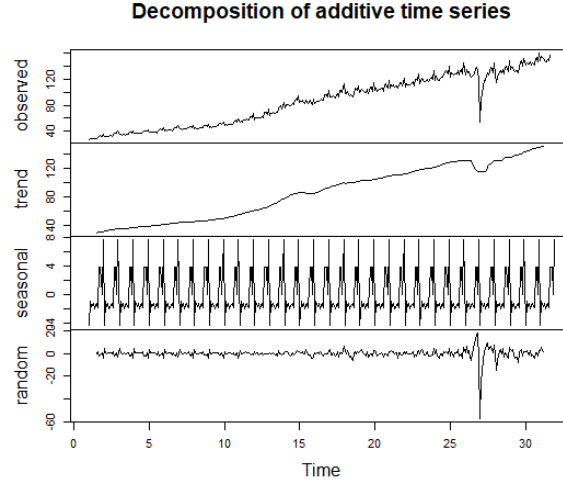


Figure 2: Decomposition into Additive Model

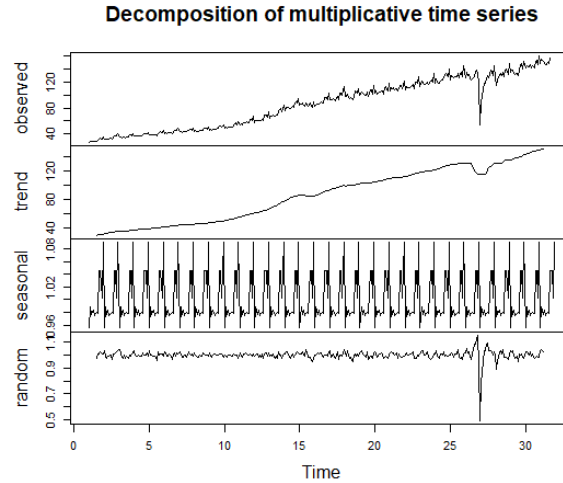


Figure 3: Decomposition into Multiplicative Model

It is expected that the model with lower variation of residuals explains the series better. Total variation may be influenced strongly by the presence of a single outlier, so we also compute the mean absolute deviation of the irregular component, which is a more robust measure of variation. We compute the total variation of the residuals and the mean absolute deviation for each model, and obtain the results as shown in Table 1.

Table 1: Model Comparison: Total Variation and Least Absolute Deviation

Model	Total Variation	Least Absolute Deviation
Additive Model	19.336	2.069
Multiplicative Model	0.001	1.000

These show that the multiplicative model is clearly better at explaining the variation in the series compared to the additive model. Therefore, for the rest of our analysis, we assume that the series follows a **multiplicative model**.

We used the `decompose` function of the `tseries` package in R to decompose the series using the described method.

4.2 Estimation of Trend

We observe an overall increasing trend, which is shown in Figure 4.

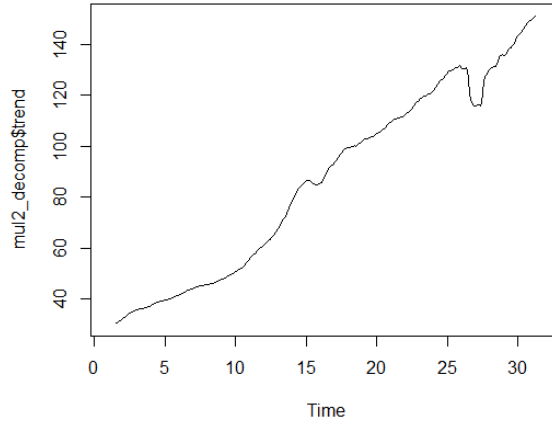


Figure 4: Estimated Trend in the Series

We obtain the trend using the method of moving averages.

4.3 Estimation of Seasonal Component

We remove the trend from our series, and then compute the seasonal component by averaging for each time unit over all periods, followed by centering. The variation of seasonal components with respect to time is as shown in Figure 5. Clearly, the seasonal components remain same with increase in level, indicating that the multiplicative model is a good fit. The seasonal indices are given in Table 2 of the Appendix.

4.4 Irregular Variation

4.4.1 Plotting Irregular Variation

After removing the trend and the seasonal component from the series, we obtain the irregular variation in the data. We consider the log-transformed residuals ($\log \epsilon_t$) since our model is multiplicative.

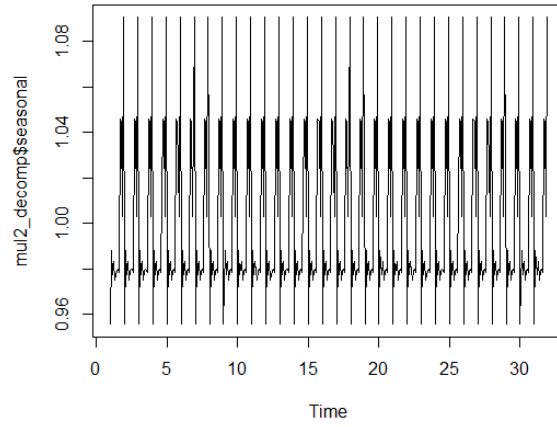


Figure 5: Variation of Seasonal Components with Time

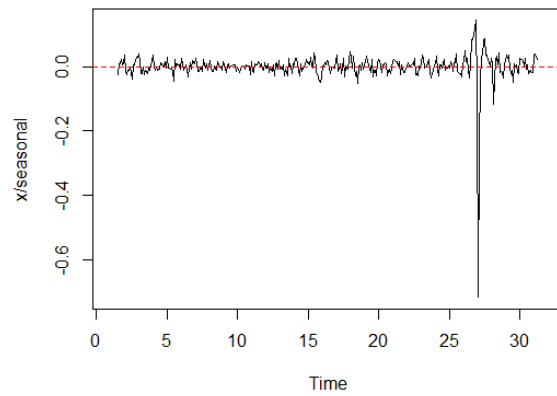


Figure 6: Variation of Irregular Components with Time

We compute the mean of the log transformed residuals to be 0.

4.5 Testing Stationarity

We perform the Augmented Dickey-Fuller test to check non-stationarity of the residuals. We obtain a p-value less than 0.01, which suggests that there is enough evidence in the data to reject the null hypothesis of non-stationarity of the residuals. Thus, the residuals are **stationary**.

4.5.1 Q-Q Plot

A Q-Q plot of the residuals (Figure 7) reveals that the errors are not normally distributed. Thus, we perform the White Test to check for heteroscedasticity. We obtain a

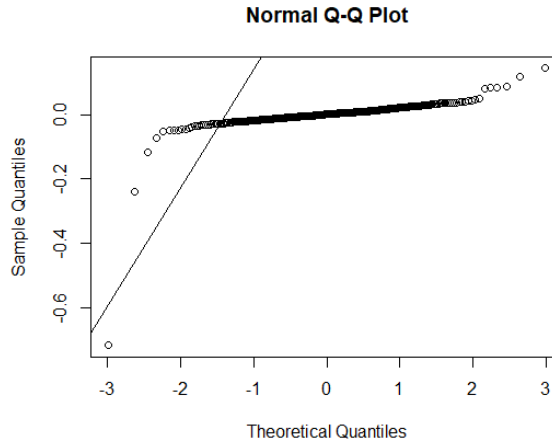


Figure 7: Q-Q Plot of Residuals

p-value of 0.1862 which indicates that there is not enough evidence in the data to reject the null hypothesis that the data is homoscedastic. Hence, it is reasonable to assume that the residuals are **homoscedastic**.

4.5.2 Testing for Autocorrelation of Residuals

The ACF plot of the residuals is shown in Figure 8. The confidence bounds are calculated as $\pm \frac{1.96}{\sqrt{n}}$ where n is the number of samples. In this time series, $n = 369$, so the confidence bounds are ± 0.1 . The plot suggests that the residuals may be white noise. We perform the Turning Point test to check if the residuals are uncorrelated. We obtain a p-value of 0.933. There is not enough evidence in the data to reject the null hypothesis of randomness of the data. Thus, it is reasonable to assume that the residuals are **independent**.

It is reasonable to assume, based on this analysis, that the residuals are **white noise**. Thus, the trend and seasonal components capture most of the variation in the data.

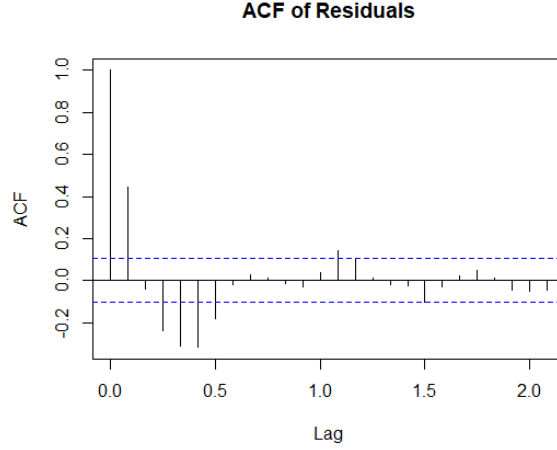


Figure 8: Autocorrelation Plot of Residuals

5 Conclusion

We have found that the combined industrial production in the sectors of mining, manufacturing and electricity has grown steadily in our country from 1994 to 2024. The COVID-19 pandemic had caused a significant dip in industrial production levels, however, the manufacturing sector has recovered from that and continues to grow. This is an encouraging sign for the Indian economy, and policymakers should adopt necessary measures to ensure the continued growth of industrial production in our country.

A Appendix

A.1 Tables

Table 2: Seasonal Indices

Month	Seasonal Index
January	1.048
February	0.999
March	1.074
April	0.969
May	0.994
June	0.977
July	0.980
August	0.975
September	0.976
October	0.985
November	0.985
December	1.035

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